



EXECUTIVE BUSINESS REVIEW / DATA ANALYTICS

STARBUCKS

Global Retail Revenue & Customer Analytics

An end-to-end analytics workflow: Excel data cleaning, a normalized PostgreSQL data model, and an executive Power BI decision layer built to answer four questions from the VP of Finance & Global Retail Operations.

EXCEL

Data cleaning & EDA

POSTGRESQL

3-table relational model

POWER BI

Model + DAX + report

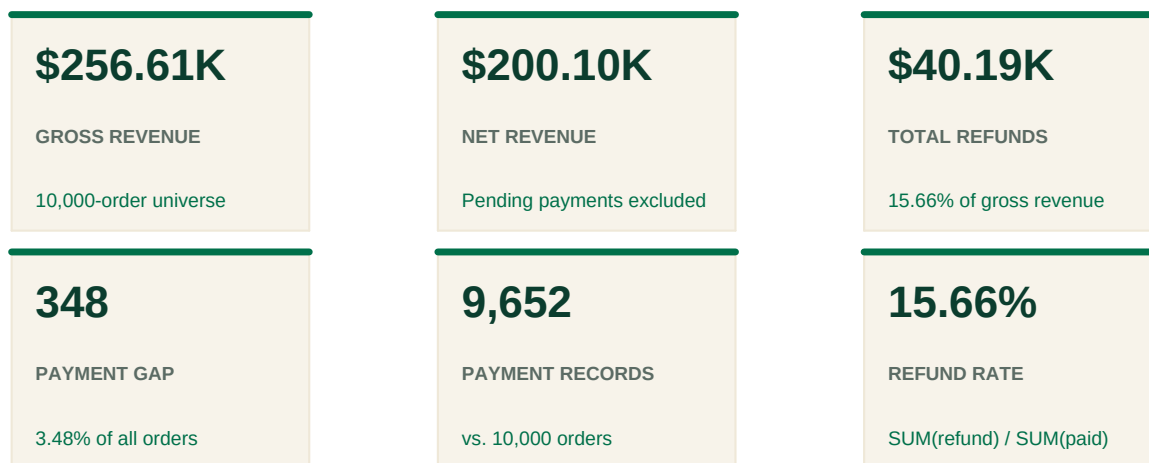
6 COUNTRIES

Drill-through views

Prepared for the VP of Finance & Global Retail Operations from the cleaned analytical dataset and the Power BI decision dashboard.

What the VP can act on

Based on 10,000 orders, 10,000 customers and 9,652 revenue records



Answering the four VP questions

Q1 · Country investment

US and China are the strongest candidates for the next round of investment. US carries the highest net revenue (\$35.37K) and the lowest refund rate (13.95%); China carries the highest gross revenue (\$44.50K) with a controlled 14.48% refund rate. Germany is high-revenue but carries the highest refund rate in the portfolio (17.54%), so scale and risk must be read together.

Q2 · Payment & refund risk

No single payment method dominates both risk types. Credit Card carries the largest payment-gap exposure (104 of 348 gap orders, 29.9%); Starbucks Card carries the smallest gap (69 orders, 19.8%) but the highest refund rate (~16.5%). Cash is the lowest-refund-rate channel (~14.7%).

Q3 · Loyalty economics

Non-Member generates the highest net revenue (\$76.56K) and the highest net revenue per customer (\$20.96), ahead of Starbucks Rewards (\$63.29K / \$19.93) and Starbucks Rewards Gold (\$60.25K / \$18.99). Programme cost, retention and incrementality are not in this dataset, so this is a revenue finding, not an ROI verdict on the loyalty programme.

Q4 · Payment-record gap

348 orders (3.48% of the 10,000-order universe) have no matching Revenue record. 214 of those are already marked Delivered — product left the store with no confirmed payment, the single highest-severity issue in the dataset. The gap is not isolated to one market: UK (66) and Canada (65) carry the largest share in the drill-through view; Japan is lowest at 43.

Start with decisions, not charts

Starbucks Corporation runs its global retail arm across six country markets (Canada, China, Germany, Japan, United Kingdom, United States), four payment channels (Cash, Credit Card, Mobile App, Starbucks Card) and three loyalty tiers (Non-Member, Starbucks Rewards, Starbucks Rewards Gold). Finance & Global Retail Operations sits on three relational tables — Customers, Orders and Revenue — but has no consolidated view across market, payment type or loyalty tier. Four questions were open before this project, and the brief was explicit: answer them with specific numbers, at the scale of ten thousand orders, not with a handful of anecdotes.

Stakeholders and what they need

Stakeholder	Role	What they need
VP of Finance & Global Retail Ops	Decision-maker	Country investment ranking; payment risk profile; loyalty programme signal
Country Market Leads	Regional owners	How their market compares to the other five on revenue, delivery and refund rate
Loyalty Programme Manager	Tier strategy	Net revenue per customer by tier
Payments & Fraud Team	Risk & compliance	Exact count of gap orders, their status, and country / category concentration
Data & Analytics Analyst (me)	Delivery	Clean data, reproducible numbers, a working dashboard, a 1-page VP summary

The four VP questions this project must answer

- Which of the six country markets deserve the next round of marketing / store investment?
- Do Cash or Starbucks Card customers carry more payment-gap or refund risk than Credit Card or Mobile App?
- Which loyalty tier delivers the best net revenue — in total and per customer?
- How large is the “order with no payment record” gap, and does it concentrate in a country or category?

Dataset at a glance

Table	Volume	Key fields
Customers.csv	10,000 rows	customer_id · country · membership_tier · signup_date
Orders.csv	10,000 rows	order_id · customer_id · product_category · payment_method · order_status
Revenue.csv	9,652 rows	order_id · amount_paid_usd · refund_amount_usd · payment_status

Why this matters

- Revenue.csv has 348 fewer rows than Orders.csv — a structural gap that is itself one of the four business problems, not a side issue to clean away quietly.

Clean first. Analyze second.

A dashboard built on uncleaned data gives confident wrong answers — worse than no dashboard at all.

1	Profile Checked row counts, field types, categorical values and key uniqueness across Customers, Orders and Revenue.
2	Standardize Normalized dates, identifiers and category labels while preserving business meaning.
3	Reconcile Compared 10,000 order records against 9,652 revenue records with a LEFT JOIN to isolate the gap.
4	Quarantine Kept 282 'Unknown' order-status rows out of delivery / attrition KPIs instead of guessing a status.
5	Separate Pending Excluded 643 'Pending' payment rows from net-revenue KPIs; tracked them as Revenue at Risk.
6	Flag & validate Built the payment-gap flag and validation checks before a single chart was drawn.

Data quality findings and how each was treated

Value	Issue	Analytical treatment
348	Revenue coverage gap	9,652 revenue rows vs. 10,000 order rows
643	Pending payments	Excluded from net revenue; tracked as revenue at risk
282	Unknown order status	Quarantined so delivery / attrition metrics stay clean
214	Delivered + no payment record	Escalated as the highest-severity exposure
15.66%	Refund rate	SUM(refund) / SUM(amount paid) — never averaged row by row
10,000 / 10,000	Order vs. customer grain	COUNT(order_id) for volume; DISTINCTCOUNT(customer_id) for reach

Senior-analyst control

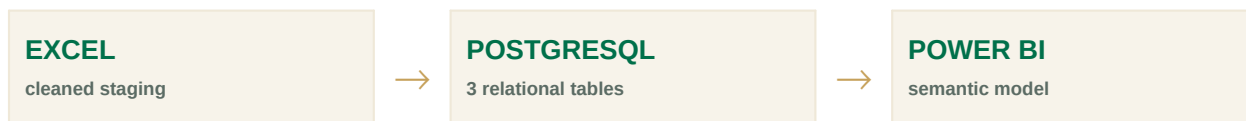
- Revenue.order_id contains duplicate values in the raw export. For a production-grade “orders with no payment record” metric, an order-level anti-join (NOT EXISTS) is used rather than assuming one revenue row per order — a simple LEFT JOIN count alone can silently over- or under-state the gap.

Excel deliverable

- Cleaned staging tables with documented decisions for every issue above.
- LEFT JOIN logic flagging the 348 gap orders, with the 214 Delivered-status subset called out separately.
- Four labeled charts: Net Revenue by Country · Payment Risk Matrix · Loyalty Tier Revenue/Customer · Gap Heatmap.
- Three-plus written business insights carried forward into the Power BI narrative.

Move from a cleaned workbook to reproducible SQL

The three case-brief tables, loaded as-is, become the queryable source of truth for every downstream KPI.



Three-table relational design (matches the case-brief grain exactly)

Table	Keys	Business columns
customers	customer_id (PK)	country, membership_tier, signup_date
orders	order_id (PK), customer_id (FK)	product_category, payment_method, order_status, order_date
revenue	order_id (FK, non-unique)	amount_paid_usd, refund_amount_usd, payment_status

Table creation & load

```

CREATE TABLE customers (
  customer_id VARCHAR(12) PRIMARY KEY,
  country VARCHAR(2) NOT NULL,
  membership_tier VARCHAR(24) NOT NULL,
  signup_date DATE
);

CREATE TABLE orders (
  order_id VARCHAR(12) PRIMARY KEY,
  customer_id VARCHAR(12) REFERENCES customers(customer_id),
  product_category VARCHAR(24) NOT NULL,
  payment_method VARCHAR(16) NOT NULL,
  order_status VARCHAR(16) NOT NULL
);

CREATE TABLE revenue (
  order_id VARCHAR(12) REFERENCES orders(order_id),
  amount_paid_usd NUMERIC(10,2) NOT NULL,
  refund_amount_usd NUMERIC(10,2) DEFAULT 0,
  payment_status VARCHAR(12) NOT NULL
);

\copy customers FROM 'customers_clean.csv' CSV HEADER;
\copy orders FROM 'orders_clean.csv' CSV HEADER;
\copy revenue FROM 'revenue_clean.csv' CSV HEADER;
  
```

Validation queries run before anything left the database

```
-- (a) revenue KPI control by payment status
SELECT payment_status, COUNT(*), SUM(amount_paid_usd), SUM(refund_amount_usd)
FROM revenue GROUP BY payment_status;

-- (b) order-level payment-gap reconciliation (anti-join, order grain)
SELECT o.order_id, o.order_status
FROM orders o
WHERE NOT EXISTS (SELECT 1 FROM revenue r WHERE r.order_id = o.order_id);
-- returns 348 rows; 214 of them have order_status = 'Delivered'

-- (c) net revenue, excluding Pending
SELECT SUM(amount_paid_usd - refund_amount_usd) AS net_revenue
FROM revenue WHERE payment_status <> 'Pending';
```

Model for reuse, then visualize

PostgreSQL tables are imported as-is and related on their natural keys before a single measure is written.

Relationships

From	Cardinality	To
customers[customer_id]	1 : *	orders[customer_id]
orders[order_id]	1 : *	revenue[order_id] (data-quality note: not always 1:1 in the raw export)

Core DAX layer (built before any visual)

Measure	DAX
Total Revenue	SUM(revenue[amount_paid_usd])
Total Refund	SUM(revenue[refund_amount_usd])
Net Revenue	CALCULATE(SUM(revenue[amount_paid_usd]) - SUM(revenue[refund_amount_usd]), revenue[payment_status] <> "Pending")
Refund Rate	DIVIDE([Total Refund], [Total Revenue])
Total Orders	DISTINCTCOUNT(orders[order_id])
Total Customers	DISTINCTCOUNT(customers[customer_id])
AOV	DIVIDE([Total Revenue], [Total Orders])
Payment Gap	CALCULATE(DISTINCTCOUNT(orders[order_id]), NOT(orders[order_id] IN VALUES(revenue[order_id])))
Payment Gap %	DIVIDE([Payment Gap], [Total Orders])

Modeling rule

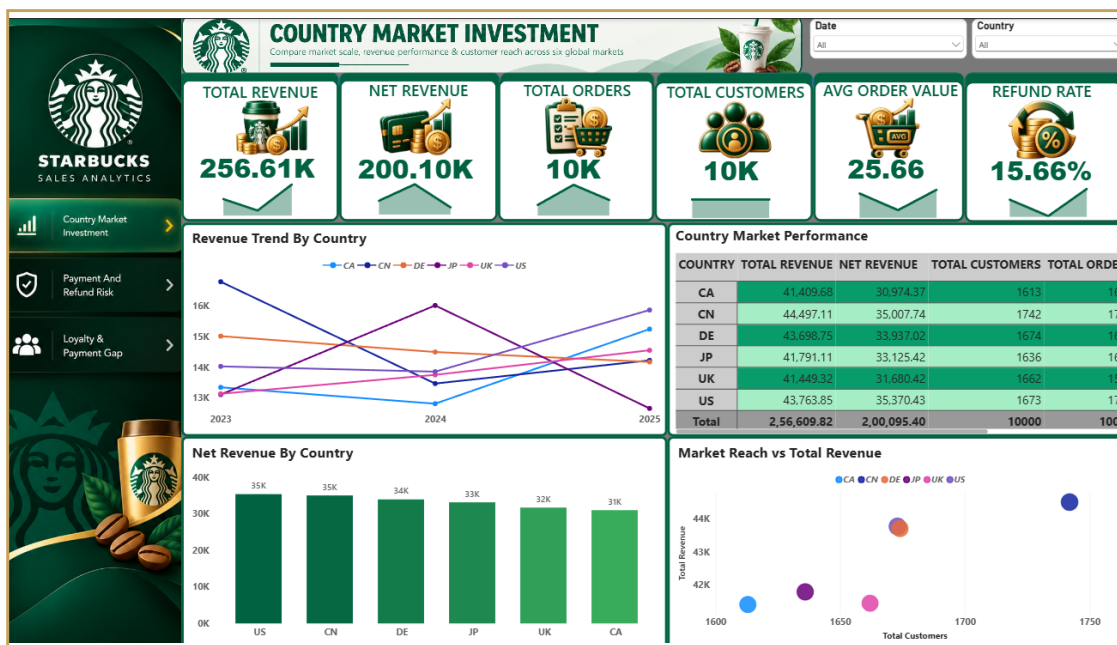
- Customer reach is always a DISTINCT customer count — never mixed with order count in the same KPI.
- Revenue and refund measures stay at revenue-record grain; Pending rows are excluded from Net Revenue by filter, not by deleting rows.

Report structure

Page	What it answers
Country Market Investment	Market scale, net economics, 2023–2025 trend, revenue-vs-reach scatter
Payment & Refund Risk	Payment gap by method, refund per method, revenue/refund/refund-rate by method, gap by country
Loyalty & Payment Gap (+ 6 country drill-throughs)	Net revenue by tier, gap orders by tier, revenue per customer; plus one drill-through page each for CA, CN, DE, JP, UK, US

Country Market Investment

From a global view to an investment lens



Live Power BI report page — Country Market Investment

Country decision table

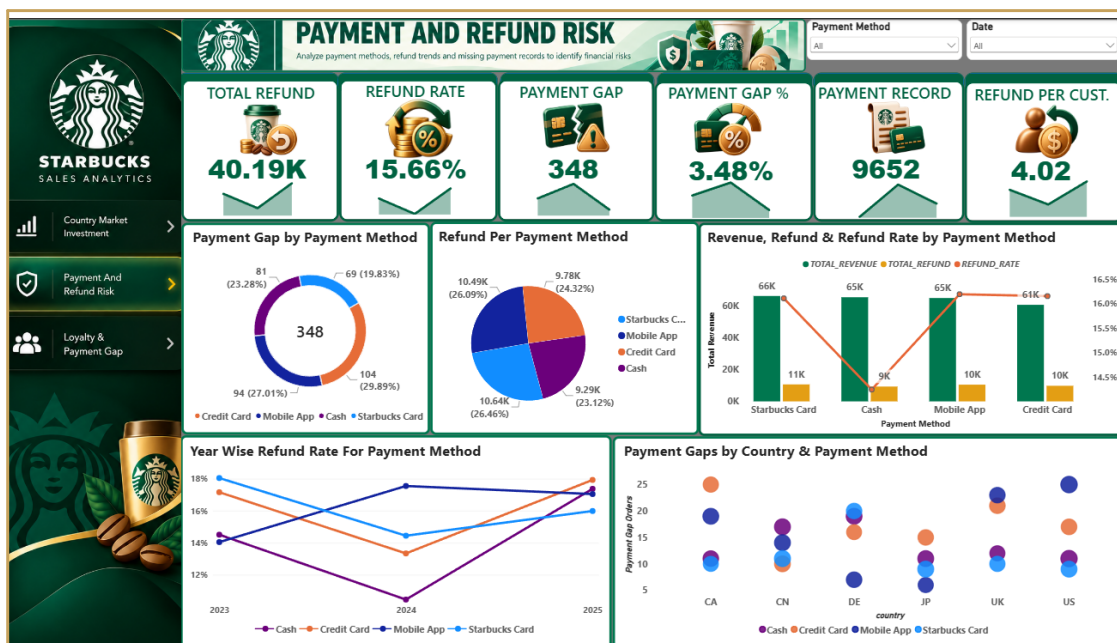
Market	Gross Rev.	Net Rev.	Refund Rate	Customers	2023 → 2025
US	\$43.76K	\$35.37K	13.95%	1,673	\$14.0K → \$15.9K
CN	\$44.50K	\$35.01K	14.48%	1,742	\$16.8K → \$14.2K
DE	\$43.70K	\$33.94K	17.54%	1,674	\$15.0K → \$14.2K
JP	\$41.79K	\$33.13K	15.22%	1,636	\$13.7K → \$12.9K
UK	\$41.45K	\$31.68K	16.09%	1,662	\$13.1K → \$14.6K
CA	\$41.41K	\$30.97K	16.79%	1,613	\$13.3K → \$15.3K

Read this first

- US has the highest net revenue and the lowest refund rate of the six markets.
- China has the highest gross revenue and the second-highest net revenue.
- Germany carries the highest refund rate (17.54%) despite strong gross revenue — scale and risk must be read together, not gross revenue alone.

Payment & Refund Risk

Reconcile cash, refunds and payment coverage



Live Power BI report page — Payment & Refund Risk

Payment method risk matrix

Payment Method	Payment Gap (share of 348)	Refund \$ (share of \$40.19K)	Refund Rate
Credit Card	104 (29.9%)	\$9.29K (23.1%)	~16.2%
Mobile App	94 (27.0%)	\$10.64K (26.5%)	~16.3%
Cash	81 (23.3%)	\$9.78K (24.3%)	~14.7%
Starbucks Card	69 (19.8%)	\$10.49K (26.1%)	~16.5%

Risk finding

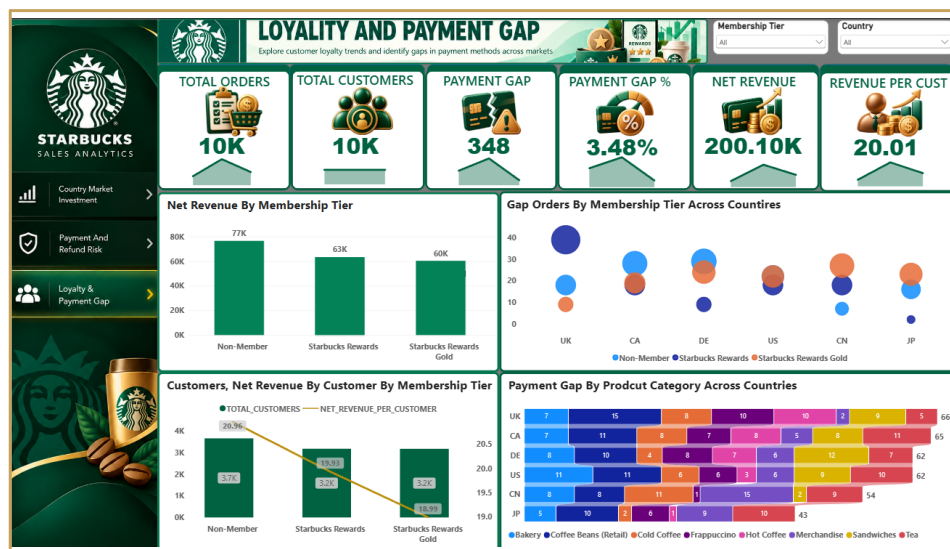
- No single payment method dominates both risk types: Credit Card leads on payment-gap count, but Cash has the lowest refund rate and Starbucks Card the highest.
- 214 of the 348 payment-gap orders are marked Delivered in the source data — product left the store with no confirmed payment. This is the single highest-severity finding in the dataset.

Control response

- Automate order-to-revenue reconciliation on a daily cadence.
- Alert immediately on any order that is Delivered with no matching payment record.
- Track Pending payments as Revenue at Risk, never as confirmed Net Revenue.
- Run payment-gap remediation and refund-rate monitoring as two separate control queues — they are different failure modes.

Loyalty & Payment Gap

Measure customer economics, not just membership volume



Live Power BI report page — Loyalty & Payment Gap

Net revenue by membership tier

Tier	Net Revenue	Customers	Net Rev. / Customer
Non-Member	\$76.56K	3,652	\$20.96
Starbucks Rewards	\$63.29K	3,175	\$19.93
Starbucks Rewards Gold	\$60.25K	3,173	\$18.99

Reported payment-gap distribution by country

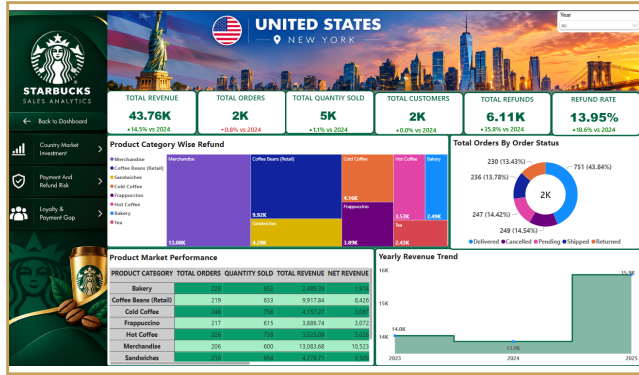
UK	CA	DE	US	CN	JP
66	65	62	62	54	43

Interpretation

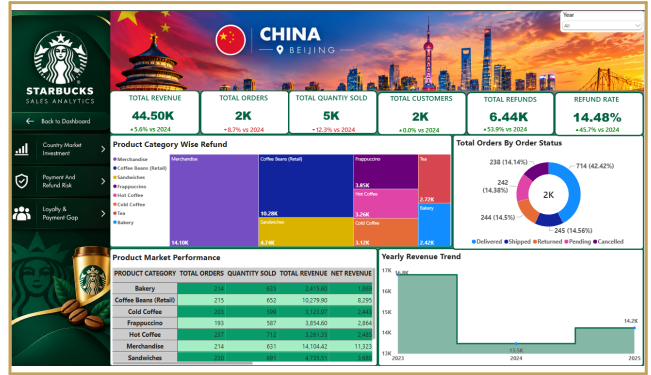
- Non-Member leads both absolute net revenue and net revenue per customer in this dataset. That does not prove the loyalty programme is under-performing — programme cost, retention lift and conversion are not present in this data and must be layered in before an ROI claim is made.
- UK (66) and Canada (65) carry the largest share of the 348-order payment gap; Japan (43) the smallest — the gap is a distributed operational issue, not a single-market defect.

Same logic, six local stories

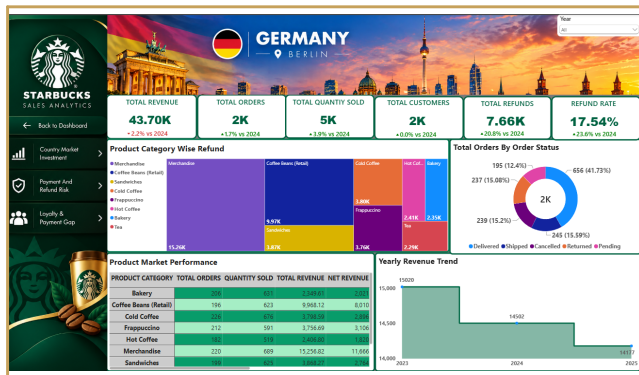
One profile page per market, filtered from the same semantic model



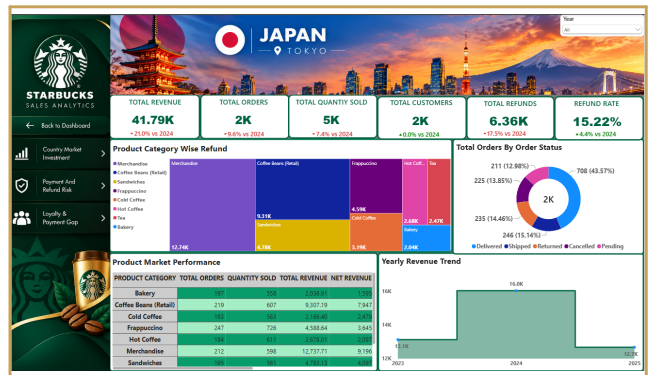
United States



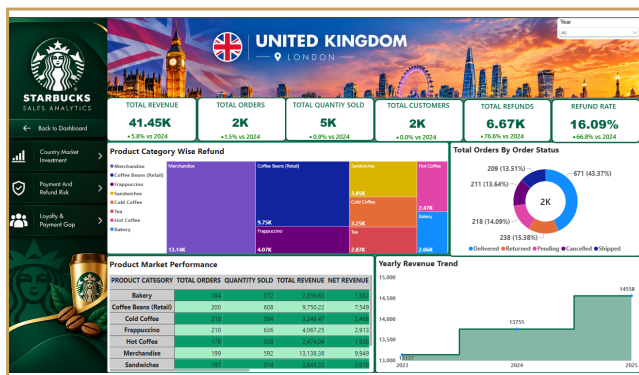
China



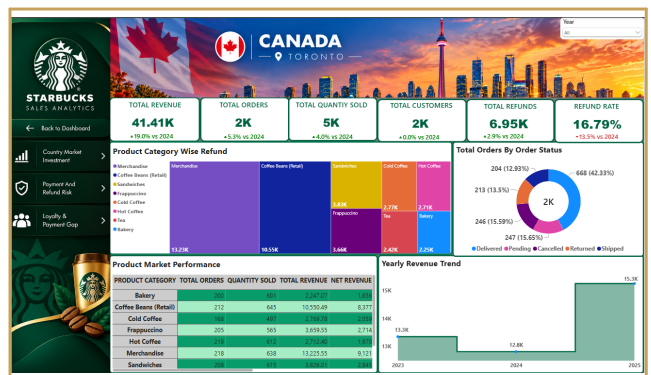
Germany



Japan



United Kingdom



Canada

Verified metrics behind each profile

Same six-market grain as the screenshots above

Market	Gross	Net	Refund	Cust.	Orders	2023 → 2025
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US — Strong net economics

\$43.76K gross | \$35.37K net | 13.95% refund. 2025 revenue reaches ~\$15.9K, the highest year-end point on the trend.

CN — Scale + efficiency

\$44.50K gross | \$35.01K net | 14.48% refund. Highest gross revenue of the six markets, though the trend cools into 2025.

DE — Refund-control focus

\$43.70K gross | \$33.94K net | 17.54% refund. Revenue declines year over year through the drill-through window.

JP — Stabilize 2025

\$41.79K gross | \$33.13K net | 15.22% refund. A 2024 peak is followed by a 2025 pullback.

UK — Protect growth, watch exposure

\$41.45K gross | \$31.68K net | 16.09% refund. Revenue rises each year, but payment-gap exposure is the highest of the six markets.

CA — Improve realization

\$41.41K gross | \$30.97K net | 16.79% refund. A 2025 rebound, but the lowest net realization among US / CN / DE / JP.

Turn charts into operating actions

01 · Investment lens

Finding: US and China combine strong net revenue with comparatively low refund rates.

Action: Allocate the next round of marketing / store investment using net economics (net revenue + refund rate), not gross sales alone.

Risk if ignored: Growth spend follows volume while refund leakage stays hidden, and Germany-style refund erosion goes unaddressed.

02 · Payment controls

Finding: Credit Card carries the highest payment-gap count (104); Starbucks Card has the lowest gap volume but the highest refund rate (~16.5%).

Action: Run two separate control queues: payment-gap remediation and refund-rate monitoring, by payment method.

Risk if ignored: A single blended "payment risk" KPI masks two different failure modes and misdirects the fix.

03 · Loyalty economics

Finding: Non-Member has the highest net revenue and net revenue per customer in the current data.

Action: Add retention, incremental revenue and programme cost to the next loyalty analysis before making any ROI claim about Rewards or Rewards Gold.

Risk if ignored: Membership value gets inferred from raw revenue volume rather than true incrementality, which can justify the wrong programme spend.

04 · Data governance

Finding: Pending payments and Unknown order statuses require explicit handling; Delivered-with-no-payment is high severity.

Action: Automate reconciliation alerts, quarantine invalid status values at the source, and publish data-quality checks alongside the dashboard.

Risk if ignored: Executives receive precise-looking numbers with hidden data-quality risk baked in.

Next iteration

- Add cost-to-serve, retention / churn, payment settlement timestamps and store-level operational metadata so the dashboard can move from descriptive reporting into performance management.